AGI ALPHA Invention Flywheel Lab

Retain the incumbent: a bounded first campaign and a verified integration repair

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Abstract

We executed a bounded first campaign toward an invention-automation enterprise whose verified capabilities enable subsequent useful work. Three materially distinct possibilities were investigated using primary evidence: grid interconnection, retired-battery qualification and industrial heat recovery. A public-network DC contingency probe passed independent equation and integrity checks, but established ordinary methods already cover its proposed numerical mechanism. A fresh successor construction matched equivalent ordinary reuse and yielded no feasible positive-transfer interval under its declared scenario. Actual reuse of Alpha's protected simulated exchange refused commercial acceptance without an independent buyer contract. A separate critic found and reproduced a real source-check ordering defect; the corrected runner refuses modified code before import. The resulting decision is to retain the incumbent and pivot before commercial admission. No complete commercial flywheel is demonstrated.

1. Possibility and strongest alternatives

The intended chain is enabling construction, accepted useful delivery, complete-cost surplus, unrestricted earned resources, further capability construction and fresh successor benefit. Repository activity and positive engineering checks cannot replace any link.

The recorded cycle allocation was 90 wall minutes with 15 included for repair and 20 reserved for verification/handoff, zero new spending, no paid provider calls, outreach or real transactions. The fixed worker is bounded to 120 CPU seconds, 150 wall seconds, 1 GiB address space and 10 MiB per file. Unavailable inference billing remains an unknown cost.

Possibility	Consequential outcome	Residual evidence gap and decision
Grid interconnection	Faster trustworthy assessment of feasible access to network capacity	Queue delays are documented, but numerical screening is not shown to cause the residual delay. Mature guarded outage methods already exist. Bounded engineering probe only.
Second-life batteries	Qualify retired batteries for useful safe service	Strong rapid-characterization predecessors exist. No rights-cleared retired-pack cohort, safety measurements or full cost baseline available. Defer.

Possibility	Consequential outcome	Residual evidence gap and decision
Industrial heat reuse	Recover otherwise wasted useful process heat	Real industrial demonstrations and competing equipment exist. No site trace, downtime, retrofit or obligations evidence available. Defer.

Lawrence Berkeley National Laboratory's 2026 queue edition supports the importance of interconnection delays [1], not the causal proposition that our numerical kernel would shorten them. DOE's roadmap identifies model, access and coordination needs [2]. PyPSA explicitly documents branch-outage distribution factors, topology dependence and bridge singularity [3]. These are contrary findings against novelty. Battery and heat sources [5-8] establish active predecessors and unresolved access requirements, not independent demand for our invention. The complete screen and dated source manifest are in `campaign/cycle-001/`.

2. Construction and unchanged specification

We retained the lab's charter and source-input screener from PR 2, preserved the owner-authored README and FIRST_CAMPAIGN bytes, and reviewed PR 1's registered-policy fixture without substituting it for the stronger existing Alpha engines. Four foundation repositories were inspected at exact source versions. Existing negative lifecycle results, generation cutoffs and unqualified concurrent candidates were preserved.

The probe loads the original PGLib v23.07 IEEE14 and IEEE30 cases [4], parsing numeric MATPOWER matrices without evaluating MATLAB. These are public AC-OPF benchmarks, not utility customer models. Our adaptation is a lossless DC reduction with branch reactance, taps, phase shifts, status, ratings and declared first-bus mismatch balancing. This balancing does not establish feasible generator dispatch.

The fixed trusted kernel applies the standard line-outage distribution factor identity. Connectivity is checked before division; an islanding outage becomes ISLANDING_UNKNOWN, never a valid delivery. A canonical model hash covers injections, electrical parameters, ratings and ordering; cached arrays have an additional integrity checksum. This resembles build system dependency invalidation, but the mapping is developer supplied and ordinary software engineering, not autonomous structural discovery.

A separate verifier assembles each connected post-outage nodal system directly from the equations [9]. It also parses the original case files independently. The frozen specification requires absolute flow agreement using atol=1e-6 MW and rtol=1e-8, matching outage and overload classifications, rejection of malformed models, source binding and exclusive evidence output. All 33 original engineering checks passed: 13 generated networks, two public networks, two parser comparisons, ten cache mutations and six malformed-input classes.

The deliberately seeded legacy bridge implementation suppresses a singular denominator and mislabels an outage CONNECTED. The correction passes the same independent specification. This transition qualifies a known safeguard; it is not a recovered historical discovery or evidence of commercial superiority.

3. A real criticism becomes an executable repair

Adversarial integration review found that the initial runner imported the candidate module before checking its frozen SHA-256. A harmless marker inserted into a temporary altered module executed before the runner raised SOURCE_IDENTITY_MISMATCH. The original runner, failed receipt and separately frozen fail-before-import requirement are preserved.

The correction verifies a hash-bound manifest of project Python sources before importing campaign, candidate, verifier and exchange modules. A further adversarial probe found that an added package initializer could escape a leaf-file manifest; the final gate therefore also refuses changes to the local Python source set, including added initializers. Public input hashes are checked before parsing. The independent replay of the altered-module probe must refuse the source without producing the marker, while the unchanged full

campaign still passes. This is an actual discovered integration defect and constructed correction, distinct from the seeded numerical ablation. It does not establish isolation against an adversary concurrently changing the shared filesystem after the check.

Local bwrap failed. No arbitrary generated strategy program was executed, and no isolation requirement was removed. The bounded trusted Python worker is not a qualified effect/network sandbox. Stronger strategy engines remain retained sources rather than being replaced by a small menu and renamed unrestricted invention.

4. Fresh successor construction and controls

The successor builds a conditional transaction envelope between the first and last IEEE30 buses. Two screening calls construct affine contingency flow responses; the constructor intersects thermal inequalities to obtain the nonnegative transfer interval, capped at 1000 MW. All arms receive the same public network and obligations.

Arm	Constructed interval	Islanding unknowns	Interpretation
Retained guarded instrument	Empty	12, 15, 33	No admissible positive transfer in the declared scenario
Fresh direct post-outage solves	Empty	12, 15, 33	Same numerical outcome
Equivalent-content ordinary reuse	Empty	12, 15, 33	Same artifact; no experience advantage
Unguarded ablation	Empty	Hidden	Unsafe information loss, not accepted improvement

The public successor was chosen before candidate outcomes. The envelope construction is development work, not protected statistical confirmation. Independent analytic tests additionally exercise positive capacity, existing-transfer headroom, an explicit cap, infeasibility and a bridge; null-versus-null agreement alone would be inadequate. Every arm explicitly denies operational admissibility because AC, voltage, dispatch, dynamic and islanding obligations remain unresolved. A faster single numerical run would not establish an advantage over equally informed ordinary reuse or pay for acquiring and maintaining the instrument.

5. Actual exchange reuse and economic consequence

We reused the two-file Alpha claim-probe ProtectedExchange closure from commit a6b2d869 (full source identity in THIRD_PARTY.md), preserving its Apache license and Git blob identities. The adapter injects its real validator and semantic checker and supplies the actual successor artifact. It does not implement another ledger or claim native Jobs/Manager settlement.

The checker binds artifact identity but refuses acceptance because no buyer-issued completed-service specification exists. Twenty mechanism checks pass, including nonpayment, rejected-output release and settlement refusal, repeated refund idempotence, duplicate refund refusal, interrupted transaction rollback/reopen, correlated refunds, persistent contract mismatch and refusal to fund a successor from startup capital. A separate role replayed the same payload and checked five adapter integrity cases. All five funded simulated offers were refused and refunded. Revenue, collections, restricted custody and unrestricted earned funds end at zero. The unchanged simulated startup cash is equity, not predecessor earnings.

No operating cohort has been admitted and no same-cohort negative-to-positive complete-cost threshold has been established. Independent demand, pricing, service obligations and complete baseline performance remain unknown. Quantity meters record numerical/exchange runtime, process peak RSS, database size and events. Inference, discovery, failed construction, verification, integration, storage, maintenance, acquisition, support and terminal costs remain unpriced rather than silently zero.

For N admitted offers, participation q, refund fraction r, price p, fixed lifecycle cost F, offer cost cO, accepted-order cost cA, incremental refund handling cR and other horizon obligations T, a conditional model is:

$$\Pi = Nq(1-r)p - F - Nc_O - Nqc_A - Nqrc_R - T.$$

For N>0, q>0 and r<1, positive surplus requires

$$p > \frac{F/N + c_O + qc_A + qrc_R + T/N}{q(1-r)}.$$

Refund principal is already netted from revenue, so it is not expensed again in cR. These symbols are not measured willingness to pay. Equal-tool or free competitors constrain any feasible price. Operator benefit, AGI ALPHA capture, investment NPV, liquidity and unrestricted earnings must remain separate findings. This cycle cannot finance its successor from earned resources.

6. Decision, reproducibility and limits

Judgment	Result
Engineering execution	Demonstrated in reduced DC and failure-boundary scope
Consequential enabling change	Not demonstrated; known incumbent retained
Commercial viability / investment attractiveness	Unknown / unknown
AGI ALPHA value capture	Unknown
Unrestricted earned funding	Not demonstrated
Causal successor benefit	Not established; ordinary reuse parity
Progressive experience advantage	Not tested
Complete connected flywheel	Not demonstrated

The decision is RETAIN_INCUMBENT and PIVOT before commercial admission. Further favorable seeds of the same commodity fixture would not change it. A new construction requires a source-bound residual constraint that competent current means do not already remove. For the grid lead the next decisive input is a rights-cleared operating model and independently specified service, measured against the full existing route. The battery and heat alternatives require physical/operational data before claims.

The one-command replay is `python reproduce.py --output /tmp/alpha-NEW` after installing the pinned requirements. Output is exclusive; incomplete runs remain preserved and recovery uses a new output path. Original raw failures and corrected runs, source closure, economic reconciliation and capability record accompany this edition. Kernel SHA-256 is 72d14ed2fe728b669e8a77834f88f7b3c6c7521537c30b616286719933874f01.

The verifier did not modify the candidate. However, builder access to part of panel source occurred before freeze; exposure was retained rather than erased by regeneration. Shared model family, libraries, tools, host and filesystem provide procedural separation, not independent external certification or OS-enforced secrecy. All results are development engineering evidence; zero new confirmation alpha is consumed and no previous claim budget is reset. No background continuation is implied.

Sources and exact evidence

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